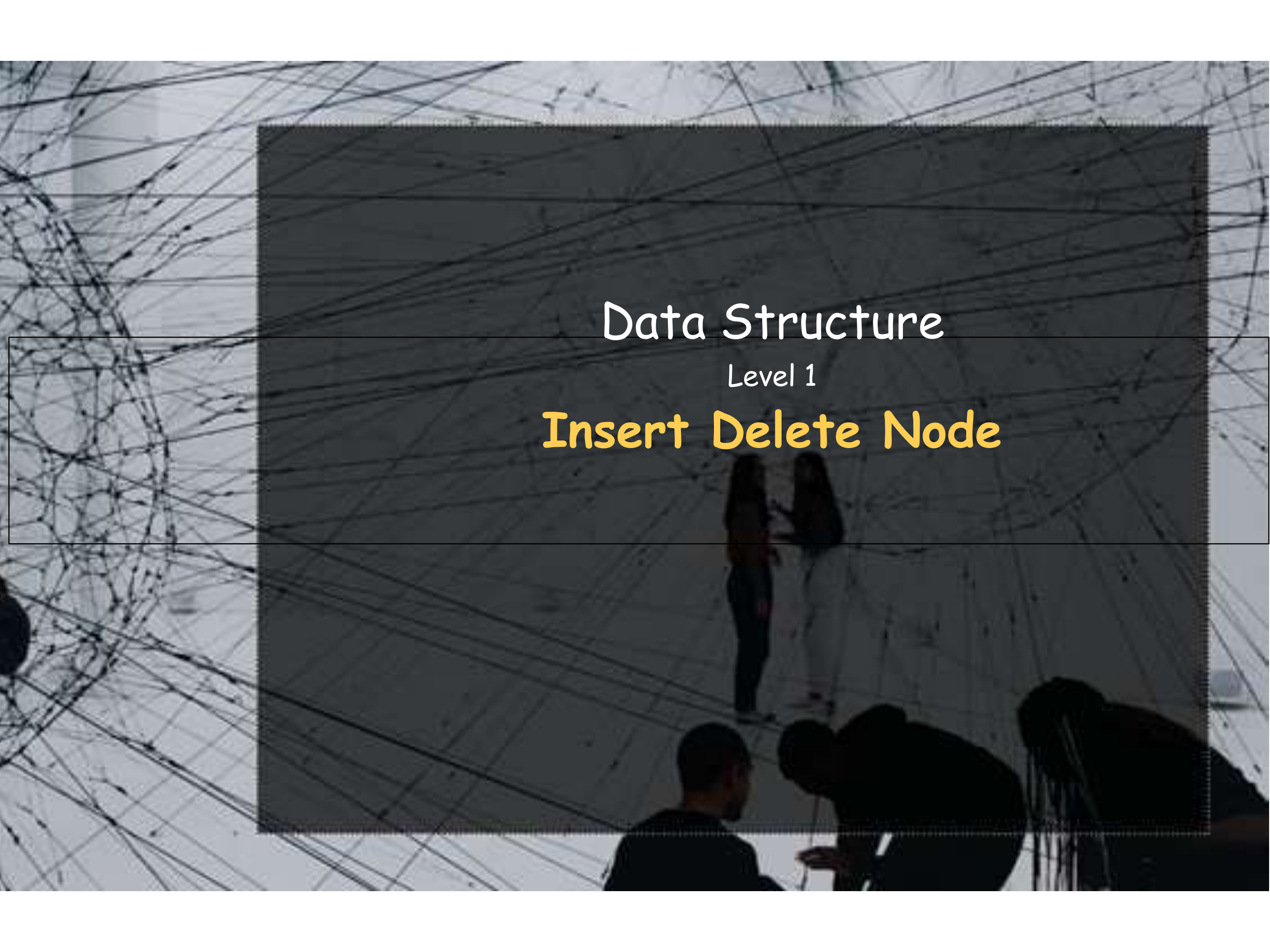


Data Structure

Level 1

Doubly Linked List Operation

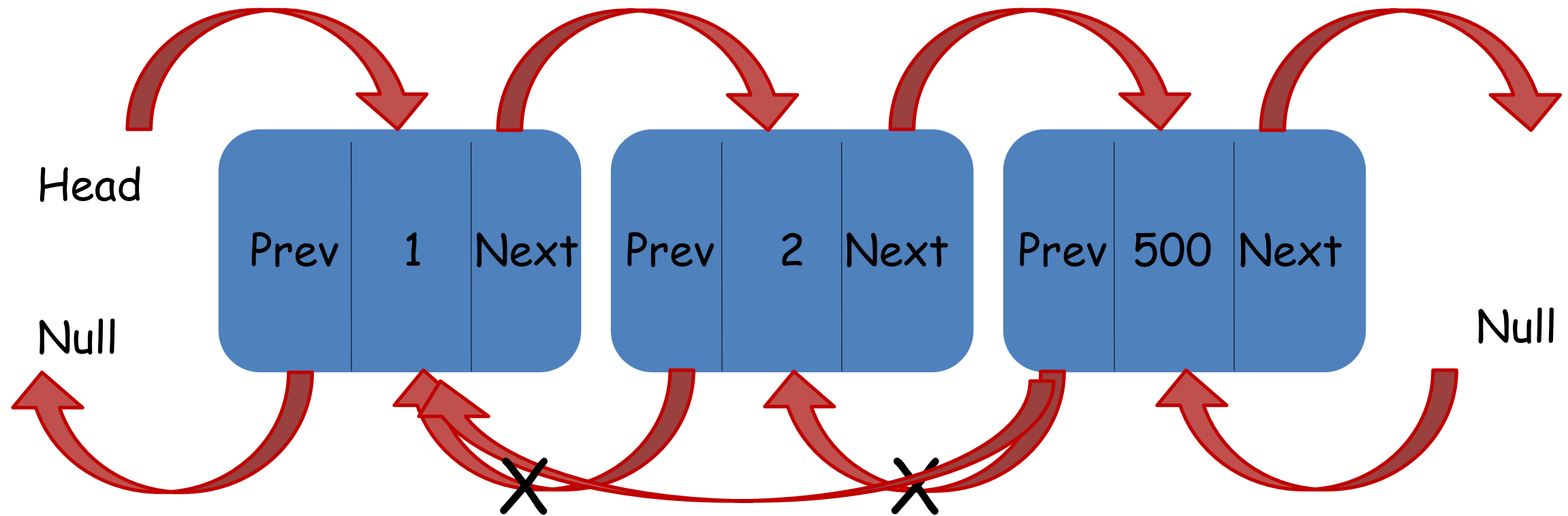
The background of the slide features a complex, abstract network of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in a discussion or collaborative work, adding a human element to the technical theme.

Data Structure

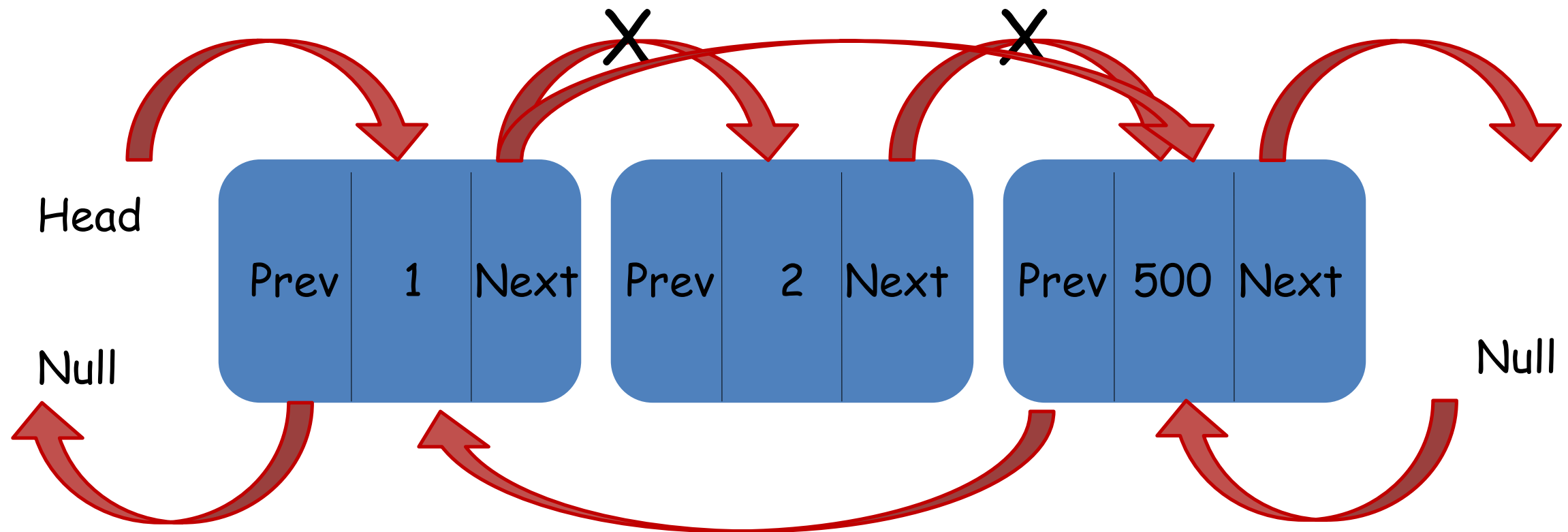
Level 1

Insert Delete Node

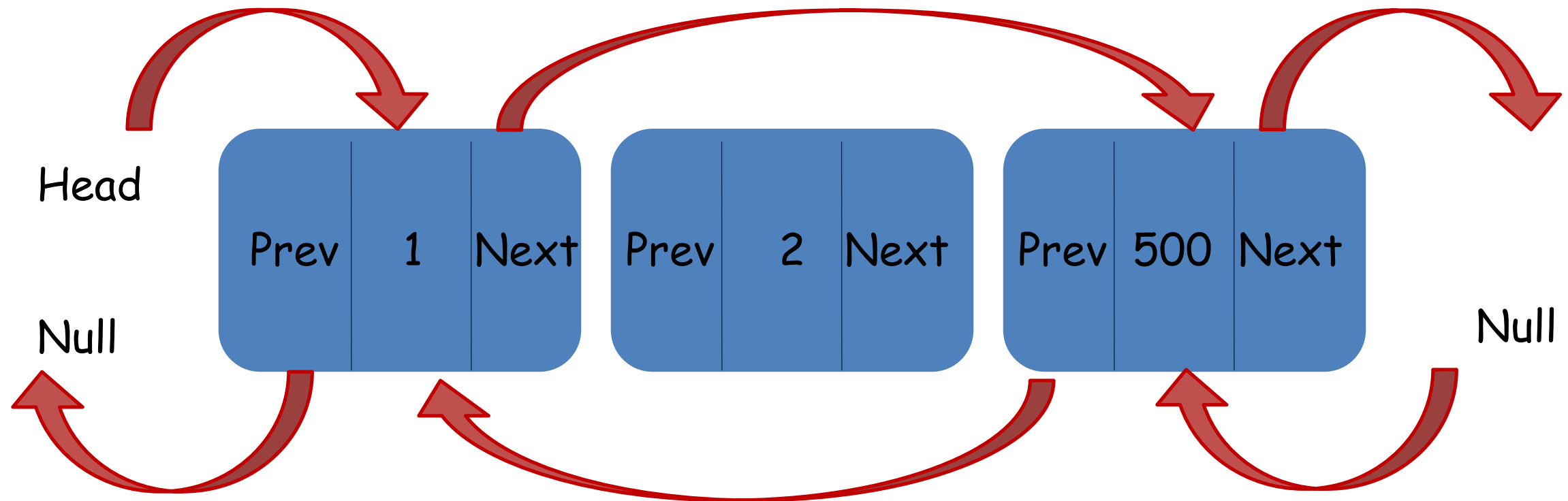
Doubley Linked List Data Structure



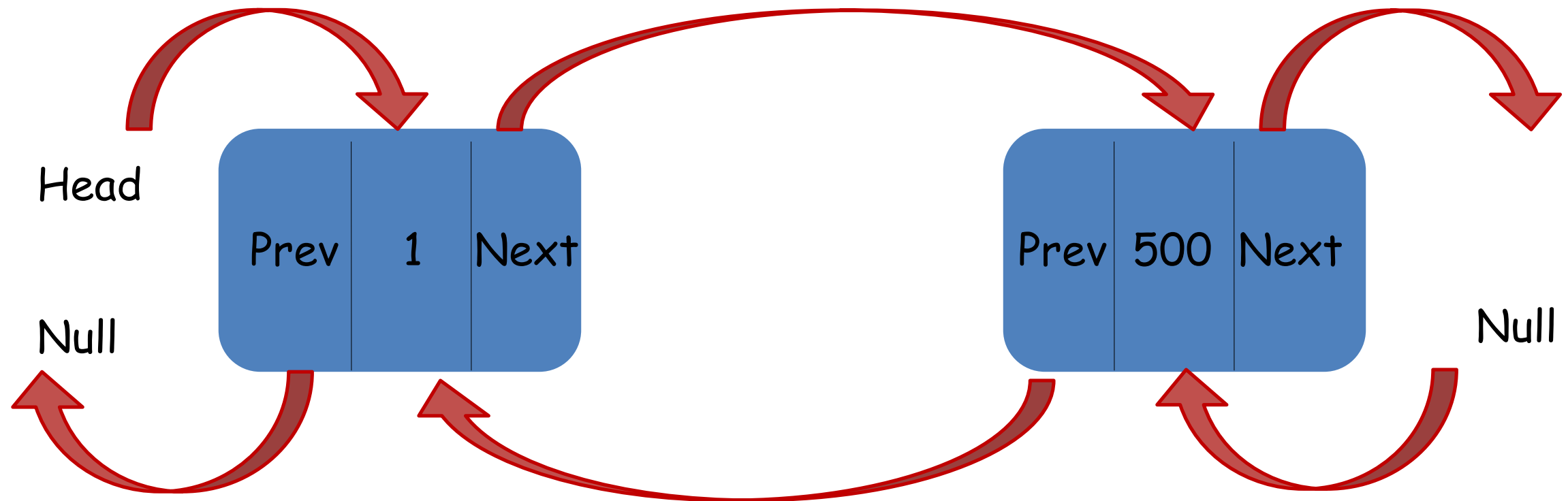
Doubley Linked List Data Structure



Doubley Linked List Data Structure



Doubley Linked List Data Structure



Definition

Removes a specific node from the doubly linked list and reconnects the surrounding nodes correctly.

Explanation

- ◆ Deletion requires updating both:
 - next pointer of the previous node
 - prev pointer of the next node
- ◆ Special handling is needed if the node is:
 - The head
 - The last node

□ Procedures

① Validate Input

□ Check:

□ If head == NULL

□ If NodeToDelete == NULL

→ □ Result: Stop if invalid

□ Procedures

2 □ If Node is Head

□ Steps:

□ Move head to next node

□ Procedures

3 □ Update Next Node

- If next exists:

- Set $\text{next} \rightarrow \text{prev} = \text{current} \rightarrow \text{prev}$

□ Procedures

4 □ Update Previous Node

- If prev exists:

- Set $\text{prev} \rightarrow \text{next} = \text{current} \rightarrow \text{next}$

□ Procedures

5 □ Delete Node

- Step:

- Free memory using delete

→ □ Result: Node removed safely

💡 Example

Before:

NULL $\leftarrow$ 1 $\leftrightarrow$ 2 $\leftrightarrow$ 3 $\leftrightarrow$ 4 $\leftrightarrow$ 5 $\rightarrow$ NULL

Delete node with value 4

After:

NULL $\leftarrow$ 1 $\leftrightarrow$ 2 $\leftrightarrow$ 3 $\leftrightarrow$ 5 $\rightarrow$ NULL

Note

- ◆ Must use Find to locate node before deletion
- ◆ Works for:
 - First node ✓
 - Middle node ✓
 - Last node ✓

⊘ Warning

- ⚠️ Deleting without updating pointers correctly → breaks the list
- ⚠️ Deleting NULL pointer → crash

⚖️ Comparison

□ Insert Operation

- ✓□ Adds new node
- ✓□ Updates surrounding pointers
- ✓□ Does not remove data

□ Delete Node

- ✓□ Time Complexity: $O(n)$ (if using Find)
- ✓□ Removes node completely
- ✓□ Requires careful pointer updates

✦✦ Summary

- ✓ DeleteNode removes a specific node from the list
- ✓ Updates both next and prev pointers
- ✓ Handles head, middle, and last cases
- ✓ Requires Find to locate the node first

Thank you 😊